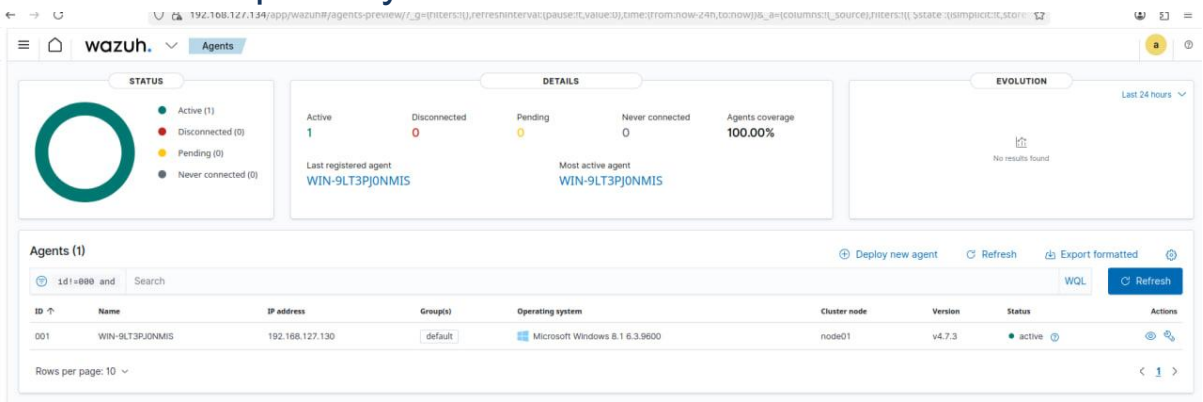
## 4. Alert Triage with Threat Intelligence

The objective of this task is to understand how a SOC analyst analyzes alerts generated by a SIEM system and verifies whether the alert is a real threat or a false positive. In this activity, alerts generated in the Wazuh were investigated and the associated indicators were validated using threat intelligence platforms such as VirusTotal and AlienVault OTX. This process helps security analysts quickly identify suspicious activity and determine whether further investigation or escalation is required.

### 1. Alert Identification

During monitoring in the Wazuh dashboard, an alert related to PowerShell execution was detected on the Windows endpoint system.



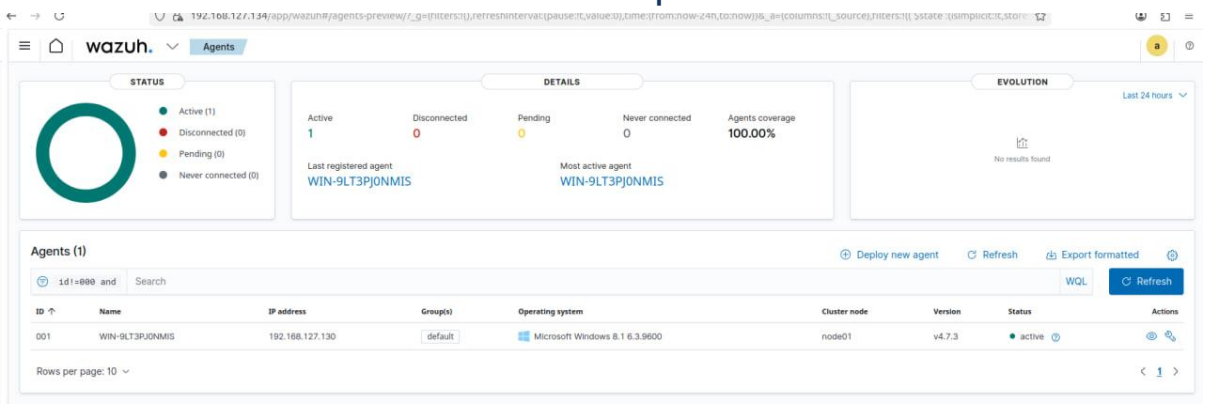

PowerShell activity is considered important in security monitoring because attackers frequently use PowerShell commands to execute scripts, download malicious payloads, or perform unauthorized system actions.
The alert generated by the SIEM system was reviewed and documented as follows.

| Alert ID | Description | Source IP | Priority | Status |
|---|---|---|---|---|
| 001 | Suspicious PowerShell Execution | 192.168.127.130 | High | Open |

This alert indicates that a PowerShell command was executed on the monitored system. Since PowerShell can be used for both legitimate administrative tasks and malicious activities, further analysis was required to understand the context of the event.

## 2. Alert Analysis

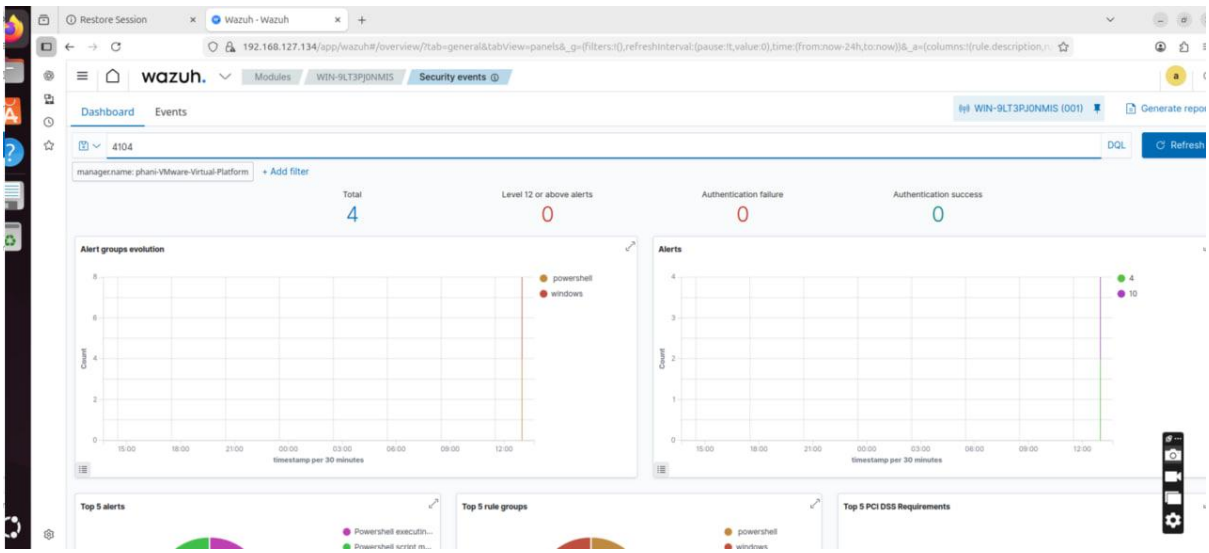
The alert was investigated in the Wazuh dashboard to check the event details and related logs. The event log showed that PowerShell commands were executed on the Windows system, which triggered the security monitoring rule. PowerShell activity is commonly associated with several attack techniques in cybersecurity, especially those related to command execution or remote script execution.




Therefore, the alert was marked as High Priority to ensure proper investigation.

During the triage process, the analyst focuses on identifying the following information:

- Source IP address generating the activity
- Command execution logs
- Associated user account
- Timestamp of the activity
- Related security events

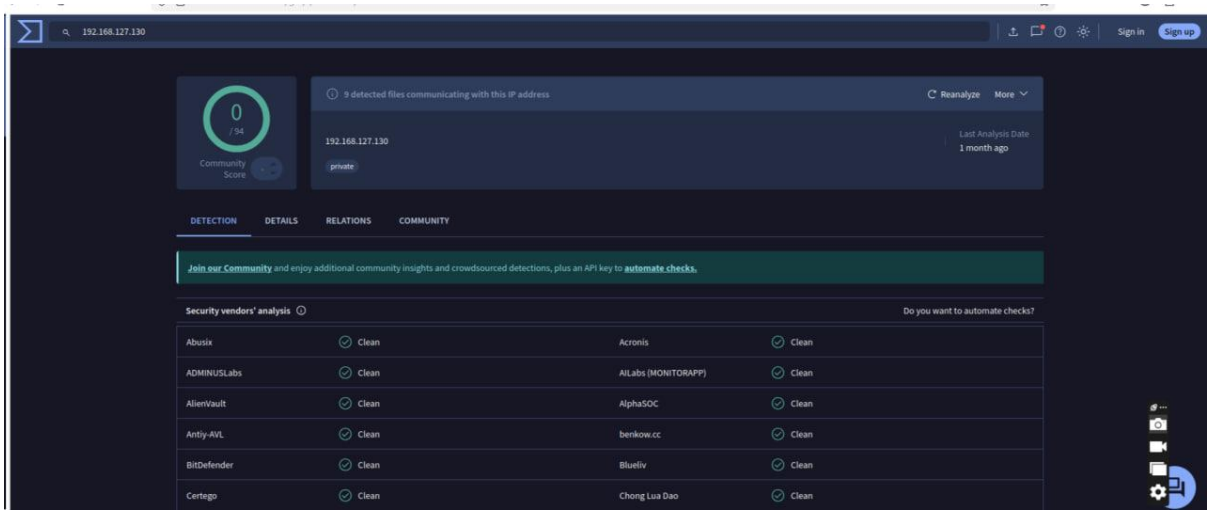



This information helps determine whether the activity is normal administrative behavior or potentially malicious activity.

## 3. Threat Intelligence Validation

To verify whether the source IP address involved in the alert was associated with known malicious activity, external threat intelligence platforms were used.

First, the IP address **192.168.127.130** was searched on VirusTotal.




VirusTotal collects threat intelligence data from multiple antivirus engines and security vendors. This helps identify whether the IP address is reported as malicious or involved in suspicious activity.

Next, the same IP address was checked in AlienVault OTX, which provides community-driven threat intelligence information about malicious indicators such as IP addresses, domains, and file hashes.
Both platforms were used to cross-reference the indicator and verify whether it is linked to known cyber threats.

## 4. Investigation Result

After performing threat intelligence validation, the IP address used in the PowerShell execution event did not appear in known malicious databases or threat intelligence feeds. Since the IP address belongs to the internal lab network environment, the activity was likely generated during testing or normal system operation.

However, because PowerShell activity can sometimes be abused by attackers, the alert was kept under observation for further monitoring.

**5. IOC Validation**

The PowerShell execution alert generated by the Wazuh SIEM system was analyzed to determine whether the activity was malicious. The associated IP address 192.168.1.101 was investigated using VirusTotal and AlienVault OTX threat intelligence platforms. The analysis did not identify any confirmed malicious indicators linked to the IP address. Based on the available evidence, the alert was considered suspicious but not confirmed as malicious. Continuous monitoring was recommended to detect any unusual PowerShell activity in the system.

**Conclusion**

Alert triage is an important process in Security Operations Center (SOC) environments because security monitoring systems often generate a large number of alerts. By analyzing alerts and validating indicators using threat intelligence sources such as VirusTotal and AlienVault OTX, analysts can quickly determine whether an alert represents a real threat or a false positive.
This activity demonstrated how SIEM alerts can be investigated and verified using external intelligence sources to improve threat detection and response capabilities.